HW 3

1.

a)

```
(a) One simulated day
------------------------------------------------
Y1  Avg waiting time in queue        18.29 min
Y2  Proportion waiting > 5 min       71.6%
Y3  Avg number in queue               3.01
Y4  Server utilization               87.6%
Y5  Customers served                    81
Y6  Final departure                492.94 min  (5:12 pm)
```

b) The number in the queue, Q(t) is a step function that only changes at events. Between two events it's constant, so the area under it for that stretch is a rectangle: (time until the next event) x (number waiting). In my code, `area_q` is calculated in each of the cases (the rectangle until the next arrival/departure is added). This value is returned and divided by the final time (T = max{480, D_N}) to find the time average number of customers in the queue.

c) For measure r with outputs $Y_{r1}, \ldots, Y_{rn}$:

Point estimate: $\bar{Y}_r = \frac{1}{n} \sum_{j=1}^{n} Y_{rj}$

Sample standard deviation: $S_r = \sqrt{\frac{1}{n-1} \sum_{j=1}^{n} (Y_{rj} - \bar{Y}_r)^2}$

Half-width: $H_r = t_{n-1,\,0.975} \, \frac{S_r}{\sqrt{n}}$

95% CI: $[\bar{Y}_r - H_r,\ \bar{Y}_r + H_r]$

d) I ran a pilot of $n_0 = 100$ replications to estimate $\bar{Y}_r$ and S_r. Since the half-width shrinks like $1/\sqrt{n}$, measure r needs about

$n_r = \left\lceil \left(\frac{1.96 \, S_r}{0.03 \, |\bar{Y}_r|} \right)^2 \right\rceil$

replications, and I took $n = \max_r n_r$.

Measure	Pilot rel. HW	n needed
Y1	15.38%	2564

Measure	Pilot rel. HW	n needed
Y2	5.67%	349
Y3	15.39%	2567
Y4	2.72%	81
Y5	2.01%	44
Y6	1.17%	15

Y1 and Y3 set the requirement (n = 2567) because they vary the most from day to day. I then ran n new replications and checked that every relative half-width was at most 3%, adding 10% more replications whenever one wasn't. Y3 was slightly over at 2567, so the final count is n = 2823.

e) Results from n = 2823 replications

Measure	Estimate	95% CI	Rel. HW	Units
Y1 Avg waiting time in queue	17.19	[16.70, 17.67]	2.82%	min
Y2 Proportion waiting > 5 min	0.624	[0.618, 0.631]	1.03%	–
Y3 Avg number in queue	2.794	[2.713, 2.876]	2.92%	customers
Y4 Server utilization	0.796	[0.792, 0.799]	0.48%	–
Y5 Customers served	80.28	[79.95, 80.61]	0.41%	customers
Y6 Final departure	501.90	[500.88, 502.92]	0.20%	min after 9 a.m. (≈ 5:22 p.m.)

2.

a) I used Welch's moving average method: 50 pilot replications starting empty, with $\bar{W}_i(w)$ plotted for four windows per ρ.

ρ	Window w	Curve levels off near	Warm-up k	Run length (completions after warm-up)
0.50	100	customer ≈ 200	500	20,000
0.75	200	customer ≈ 800	1,500	50,000
0.95	1000	customer ≈ 3,000	5,000	200,000

Choosing w: the smallest window that gave a reasonably smooth curve without flattening the initial rise.

Choosing k: the point where the curve stops rising, rounded up for safety.

Choosing run length: at least 20 times the warmup. It also has to be long enough to keep the 95% CI within about 1-2.5% of the estimate.

b) Both the warmup and the run length grow much faster than ρ does.

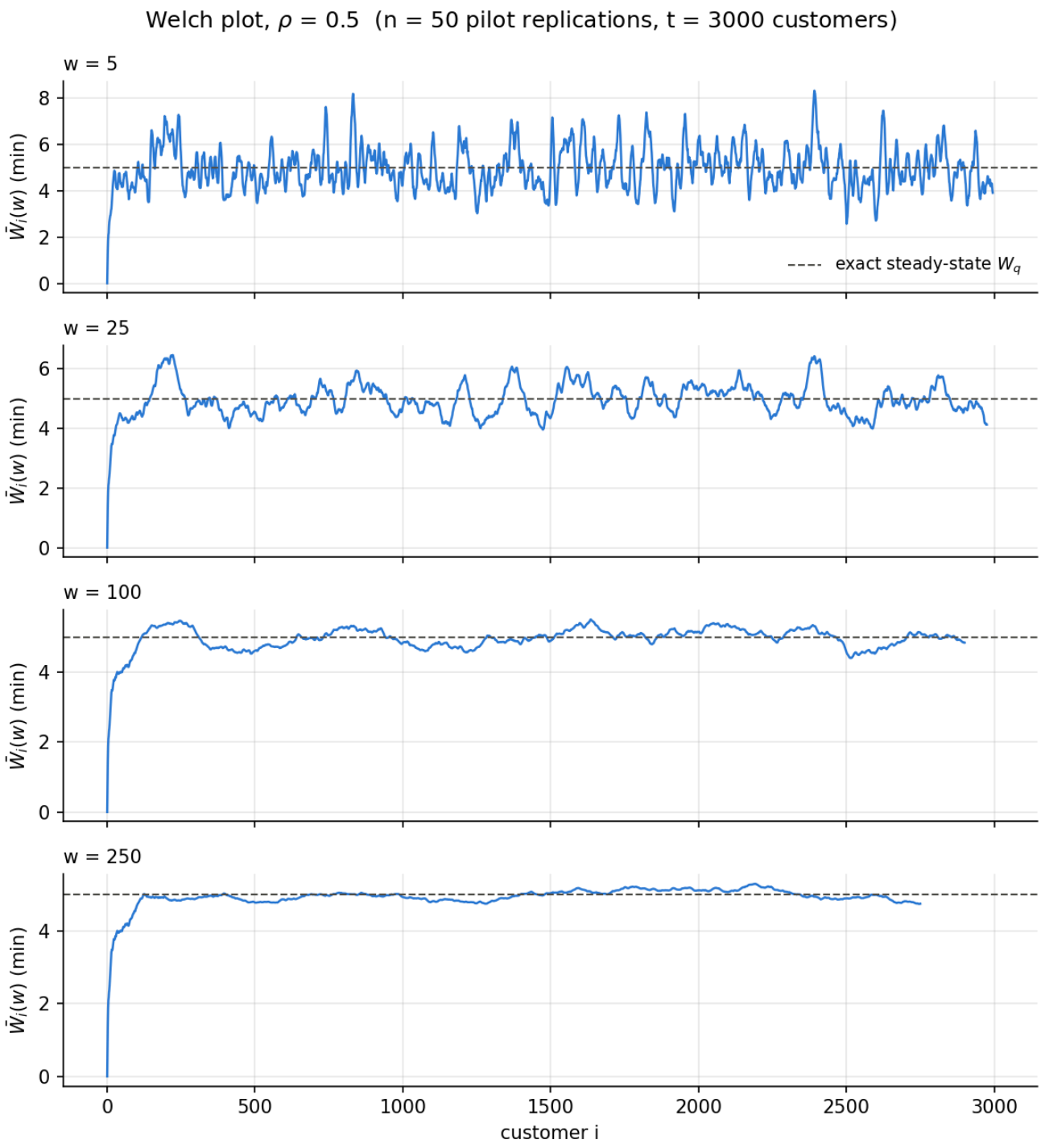
- warmup: the transient lasts about 200, 800, and 3000 customers. As $\rho \to 1$, the queue drains slowly so it takes much longer to forget the empty start.
- run length: successive waits become strongly correlated and their variance grows, so each run gives less information per customer.

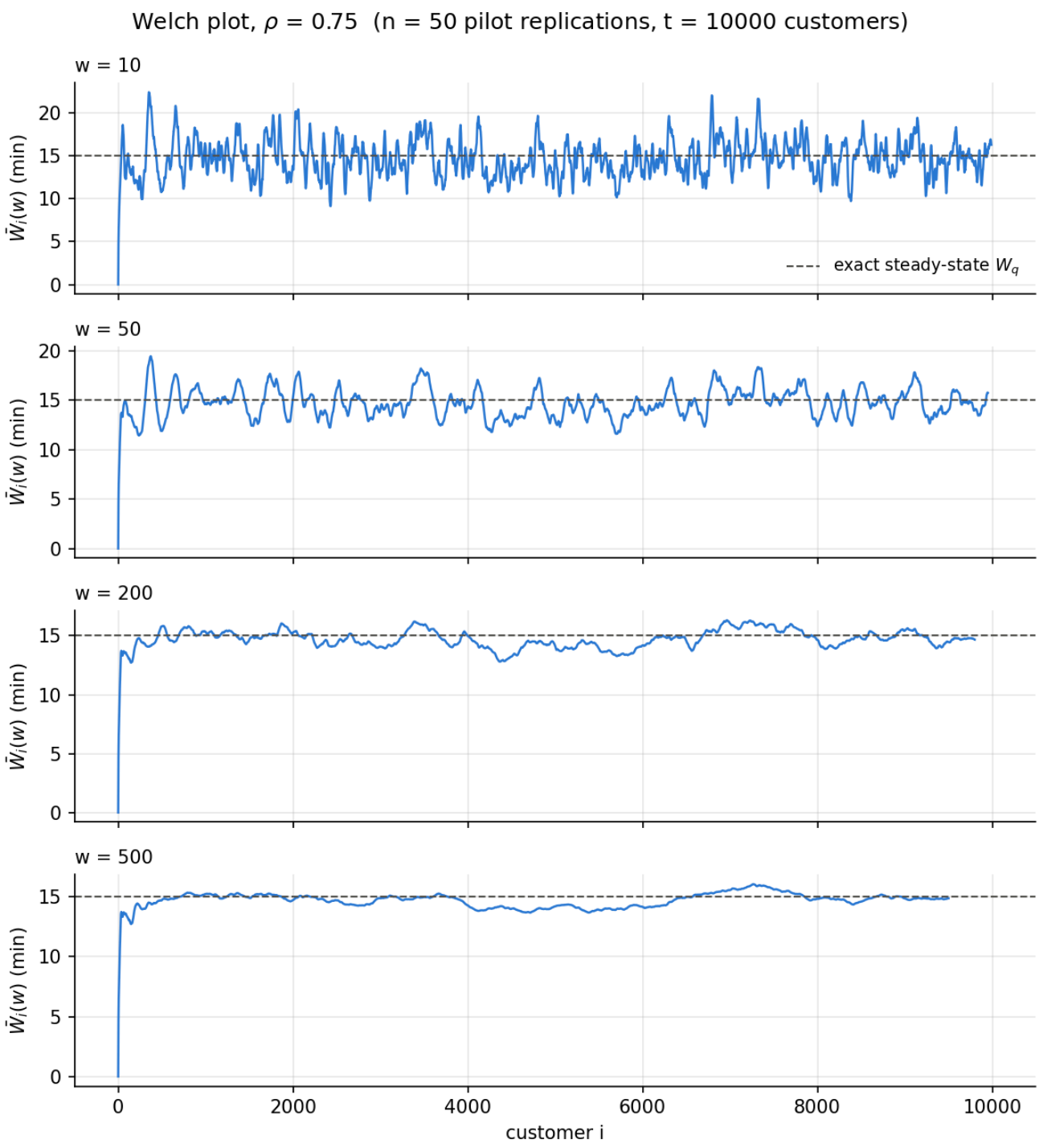
c) Results (n=50, warmup discarded)

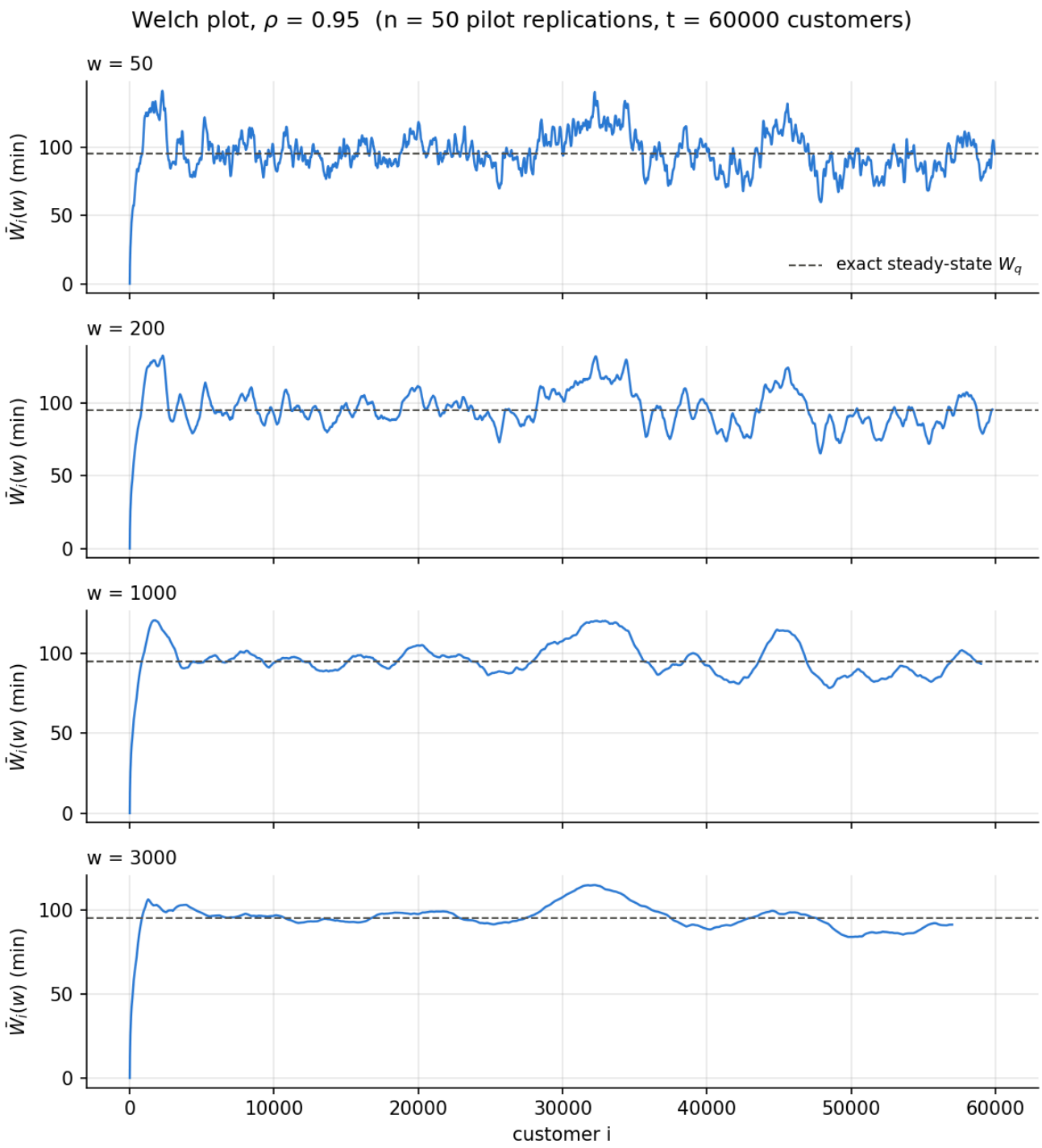
In each replication, the statistics are reset at the k-th service completion. W_q and $P(W_q > 5)$ count only customers who start service after the warm-up. L_q and U divide by the time after the warm-up.

ρ	Wq (min)	P(Wq > 5)	Lq (customers)	U
0.50	5.01 [4.95, 5.06]	0.303 [0.301, 0.305]	0.500 [0.494, 0.507]	0.500 [0.499, 0.502]
0.75	14.96 [14.81, 15.11]	0.584 [0.582, 0.586]	2.24 [2.22, 2.27]	0.750 [0.749, 0.751]
0.95	95.4 [93.1, 97.7]	0.903 [0.902, 0.905]	18.13 [17.70, 18.56]	0.950 [0.949, 0.951]

Every CI contains the exact M/M/1 value: W_q = 5, 15, 95 min; $P(W_q > 5)$ = 0.303, 0.584, 0.904; L_q = 0.5, 2.25, 18.05; U = ρ.


Welch plot, ρ = 0.5 (n = 50 pilot replications, t = 3000 customers)
w = 5
w = 25
w = 100
w = 250
$\bar{W}_i(w)$ (min)
exact steady-state W_q
customer i
0
500
1000
1500
2000
2500
3000


Welch plot, ρ = 0.75 (n = 50 pilot replications, t = 10000 customers)
w = 10
w = 50
w = 200
w = 500
$\bar{W}_i(w)$ (min)
exact steady-state W_q
customer i
0
2000
4000
6000
8000
10000
0
5
10
15
20


Welch plot, ρ = 0.95 (n = 50 pilot replications, t = 60000 customers)
w = 50
w = 200
w = 1000
w = 3000
$\bar{W}_i(w)$ (min)
0
50
100
exact steady-state W_q
0
10000
20000
30000
40000
50000
60000
customer i